Algebraic Notation and Substitution

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KEY WORDS expression *(kīanga taurangi)* · variable *(taurangi)* · substitute *(whakakapi)* · formula *(tātai)*

A Algebraic notation ●○○

Remember: In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $3 \times a$ = _____ **b)** $b \times 5$ = _____ **c)** $a \times b$ = _____ **d)** $a \times a$ = _____

e) $x \div 4$ = _____ **f)** $2 \times a \times b$ = _____ **g)** $y \times y \times y$ = _____ **h)** $(a + 3) \times 2$ = _____

i) $4 \times x \times x$ = _____ **j)** $(x + y) \div 2$ = _____ **k)** $1 \times n$ = _____ **l)** $3 \times a + 2 \times b$ = _____

B Writing expressions ●○○

Watch out: "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 5 more than n _____ **b)** n multiplied by 4 _____

c) 6 less than n _____ **d)** n divided by 3 _____

e) double n, then add 1 _____ **f)** n squared _____

g) add 2 to n, then multiply by 5 _____ **h)** subtract n from 10 _____

C Substituting into an expression ●●○

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

3. Find the value of each expression when $a = 3$ and $b = 5$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $4(b - a)$ = _____ **h)** $10 - 2a$ = _____

i) $b^2 - a^2$ = _____ **j)** $2ab$ = _____ **k)** $\frac{6b}{a}$ = _____ **l)** $\frac{a + b}{2}$ = _____

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D Negative numbers and decimals ●●●

⚠ **Watch out:** Put brackets around a negative number when you substitute it: if $x = -2$, then $x^2 = (-2)^2 = 4$.

4. Find the value of each expression when $x = -2$ and $y = 4$.

a) $x + y$ **b)** $3x$ **c)** xy **d)** $y - x$

e) x^2 **f)** $2x + y$ **g)** $5 - x$ **h)** $\frac{y}{x}$

5. Find the value of each expression when $p = 0.5$ and $q = 6$.

a) $2p$ **b)** pq **c)** $p + q$ **d)** $q - 4p$

E Substituting into a formula ●●●

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 8$ and $w = 5$.

b) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 7$ and $w = 4.5$.

c) The cost in dollars of hiring a hall for n hours is $C = 5n + 20$. Find C when $n = 6$.

d) The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 6$ and $h = 4$.

e) The speed of a car is $v = u + at$. Find v when $u = 5$, $a = 3$ and $t = 4$.

f) The temperature on a mountain h kilometres up is $T = 15 - 2h$. Find T when $h = 9$.

★ Extension

7. Which is larger when $x = 3$: $2x$ or x^2? Which is larger when $x = 1$?

8. Try different values of n to find the one that makes $3n + 1 = 16$ true.

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Algebraic Notation and Substitution

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS expression *(kīanga taurangi)* · variable *(taurangi)* · substitute *(whakakapi)* · formula *(tātai)*

A Algebraic notation ●○○

Remember: In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $5 \times a$ = _____ **b)** $c \times 7$ = _____ **c)** $x \times y$ = _____ **d)** $b \times b$ = _____

e) $y \div 3$ = _____ **f)** $4 \times p \times q$ = _____ **g)** $a \times a \times a$ = _____ **h)** $(b + 2) \times 5$ = _____

i) $3 \times y \times y$ = _____ **j)** $(a + b) \div 4$ = _____ **k)** $1 \times m$ = _____ **l)** $2 \times x + 5 \times y$ = _____

B Writing expressions ●○○

Watch out: "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 8 more than n _____ **b)** n multiplied by 6 _____

c) 4 less than n _____ **d)** n divided by 5 _____

e) triple n, then subtract 2 _____ **f)** n cubed _____

g) add 4 to n, then multiply by 3 _____ **h)** subtract n from 20 _____

C Substituting into an expression ●●○

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

3. Find the value of each expression when $a = 4$ and $b = 6$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $4(b - a)$ = _____ **h)** $10 - 2a$ = _____

i) $b^2 - a^2$ = _____ **j)** $2ab$ = _____ **k)** $\frac{6b}{a}$ = _____ **l)** $\frac{a + b}{2}$ = _____

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D Negative numbers and decimals ●●●

⚠ **Watch out:** Put brackets around a negative number when you substitute it: if $x = -2$, then $x^2 = (-2)^2 = 4$.

4. Find the value of each expression when $x = -3$ and $y = 6$.

a) $x + y$ **b)** $3x$ **c)** xy **d)** $y - x$

e) x^2 **f)** $2x + y$ **g)** $5 - x$ **h)** $\frac{y}{x}$

5. Find the value of each expression when $p = 1.5$ and $q = 4$.

a) $2p$ **b)** pq **c)** $p + q$ **d)** $q - 4p$

E Substituting into a formula ●●●

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 9$ and $w = 6$.

b) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 8$ and $w = 3.5$.

c) The cost in dollars of hiring a boat for n hours is $C = 8n + 15$. Find C when $n = 5$.

d) The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.

e) The speed of a car is $v = u + at$. Find v when $u = 4$, $a = 2$ and $t = 6$.

f) The temperature on a mountain h kilometres up is $T = 12 - 2h$. Find T when $h = 8$.

★ Extension

7. Which is larger when $x = 4$: $2x$ or x^2? Which is larger when $x = 1.5$?

8. Try different values of n to find the one that makes $4n - 3 = 17$ true.

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